10. 10. 2025

1. π_1 (Supplier)

2. π_2 (Part)

3. $\sigma_{City = 'London'}$ (Supplier)

4. $\pi_{PNO, PName, Color}$ ($\sigma_{City = 'Paris'}$ (Part)

5. $\pi_{PNO, PName}$ ($\sigma_{Weight > 15}$ (Part))

6. $\pi_{PNO, PName}$ ($\sigma_{Weight > 15 \wedge Color \neq 'Red'}$ (Part))

7. $\pi_{PNO, PName}$ ($\sigma_{Weight > 15 \wedge Color \neq 'Red' \wedge Color \neq 'Green'}$ (Part))

8. $\pi_{PNO, PName}$ ($\sigma_{Weight >= 15 \wedge Weight <= 19}$ (Part))

9. $\pi_{P.PNO, PName, Color, Weight, City}$ (Part $\overset{PNO}{\bowtie}$ ($\sigma_{SNO = 'S1'}$ (Shipment))

10. $\pi_{Sh.SNO, SName, Status, City}$ (($\sigma_{PNO = 'P1'}$ (Shipment) $\overset{SNO}{\bowtie}$ Supplier)

11. $\pi_{S.SNO, SName, Status, City}$ (($\sigma_{City = 'London'}$ (Supplier) $\overset{SNO}{\bowtie}$ Shipment) $\overset{PNO}{\bowtie}$ $\sigma_{City = 'London}$ (Part))

12. $\pi_{P.PNO, PName, Color, Weight, City}$ π_{PNO} ($\sigma_{SNO = 'S1'}$ (Shipment))

18. $A \leftarrow \pi_{Suppiler.SNO, SName, Status, City}$ (Supplier $\overset{SNO}{\bowtie}$ Shipment)

$B \leftarrow \pi_{Suppiler.SNO, SName, Status, City}$

$C \leftarrow B - A$

20 $\pi_{Suppiler.SNO, SName, Status, City}$ ($\sigma_{SNO = NULL}$ (Supplier $\overset{SNO}{\bowtie}$ Shipment)

21. $\Im_{count (SNO)}$ (Suplier)

22. $\Im_{Count (SNO)}$ ($\sigma_{City = 'London'}$ (Supplier))

23. $\Im_{Max(Status), Min(Status)}$ (Supplier)

24. $\Im_{Max(Status), Min(Status)}$ ($\sigma_{City = 'London'}$ (Supplier))

25. $_{SNO}\mathfrak{I}_{sum(Qty)}$ (Shipment)

26. $_{SNO,\ SName,\ City}\mathfrak{I}_{sum(Qty)}$ (Supplier $\overset{SNO}{\bowtie}$ Shipment)

27. $_{SNO,\ SName,\ City}\mathfrak{I}_{sum(Qty)}$ (Supplier $\overset{SNO}{\bowtie}$ Shipment) $\leftarrow$ A

π_{SNO} ($\sigma_{sum(Qty) > 500}$ (A))

28. $_{SNO}\mathfrak{I}_{sum(Qty)}$ ($\sigma_{Part.color = 'Red'}$ (Shipment $\overset{PNO}{\bowtie}$ Part)) $\leftarrow$ A

π_{SNO} ($\sigma_{sum(Qty) > 500}$ (A))

29. $\sigma_{Part.Color = 'Red'}$ ((Supplier $\overset{SNO}{\bowtie}$ Shipment) $\overset{PNO}{\bowtie}$ Part) $\leftarrow$ A

$_{SNO,\ SName,\ City}\mathfrak{I}_{sum(Qty)}$ (A) $\leftarrow$ B

$\pi_{SNO,\ SName,\ City,\ SUM(Qty)}$ ($\sigma_{SUM(Qty) > 300}$ (B))

30. $_{City}\mathfrak{I}_{COUNT(*)}$ (Supplier)

31. $_{SNO,\ SName}\mathfrak{I}_{sum(Qty)}$ (Supplier $\overset{SNO}{\bowtie}$ Shipment) $\leftarrow$ A

$\mathfrak{I}_{Max(Sum(Qty))}$ (A) $\leftarrow$ B

$\pi_{SNO,\ SName}$ ($\sigma_{sum(Qty) = Max(Sum(Qty))}$ (B))

32. π_{City} (Supplier) $\cup$ π_{City} (Part)